Fetal Health Classification from Cardiotocography Data

Machine Learning Mini-Project | One-page technical summary

Problem statement. Cardiotocography (CTG) records fetal heart-rate and uterine-activity patterns. The objective was to build and compare supervised learning models that classify each recording into three fetal-health categories: Normal, Suspect, or Pathological. Because the classes are imbalanced, the study emphasizes class-balanced Macro F1 in addition to accuracy.

Dataset details. The project uses the UCI Cardiotocography dataset (2,126 original observations). Inspection found no missing values; 13 duplicate rows were removed, leaving 2,113 unique cases and 21 numeric CTG features (including baseline heart rate, accelerations, decelerations, variability, histogram statistics, and uterine contractions). Class counts are Normal 1,646 (77.90%), Suspect 292 (13.82%), and Pathological 175 (8.28%). Data were split stratified 80:20 into 1,690 training and 423 test rows. The stored integer target IDs are 0/1/2; the project documentation notes that the semantic mapping is inferred from the source class counts.

Approach. Exploratory analysis used descriptive summaries, class counts, feature distributions, and correlation inspection. Four classifiers were compared: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Model selection used 5x3 repeated stratified cross-validation on the training set, with Macro F1 as the primary metric. The strongest model, XGBoost, was tuned with GridSearchCV using the training data only. The untouched test set was used once for final metrics, and a percentile bootstrap interval was computed for test Macro F1. A separate Random-Forest-based feature-selection experiment compared all 21 features with the top 10.

Implementation overview. The reproducible workflow is implemented in the Jupyter notebooks *01_data_analysis.ipynb* and *02_Model_Training.ipynb*. Processed train/test CSV files, selected features, trained pickle models, metadata, evaluation tables, and plots are stored under *data/processed*, *models*, and *results*. Reusable evaluation helpers calculate accuracy, weighted precision/recall/F1, Macro F1, per-class metrics, bootstrap confidence intervals, and confusion matrices.

Results. Comparative performance on the held-out test set:

Model	CV Macro F1	Test Accuracy	Test Macro F1
Logistic Regression	0.7802	0.9362	0.8745
Decision Tree	0.8547	0.9078	0.8389
Random Forest	0.8808	0.9669	0.9383
XGBoost	0.8969	0.9645	0.9358
XGBoost (tuned)	**0.9005**	**0.9716**	**0.9511**

Conclusions. Tuned XGBoost using all 21 features was the best overall model (learning rate 0.1, depth 3, min child weight 1, 200 estimators), reaching 0.9716 accuracy and 0.9511 test Macro F1 (95% bootstrap CI: 0.9205–0.9765). Its cross-validated Macro F1 of 0.9005 is the more conservative generalization estimate. Reducing to 10 features lowered test Macro F1 to 0.9379, so the full feature set is recommended when performance is the priority. The main limitation is the small minority-class test support; nine Suspect cases were predicted as Normal, so future work should consider calibration, threshold tuning, and validation on additional clinical data. This is an educational research project, not a medical device or a basis for clinical decisions.